

21.1 Checking normality

Normality turns the first two moments of least squares into exact distributions. How much of Part III needs it? [Chapter 19](#) answered the question for coefficients. When no observation carries much leverage, $\hat{\boldsymbol{\beta}}$ is approximately normal whatever the error law, and the t and F tests keep their nominal levels in large samples (Theorem 19.3.2). Two kinds of inference are not protected in this way. The interval for σ^2 of [Section 12.1](#) depends on the fourth moment of the errors (Proposition 7.5.4). A prediction interval is a statement about a single new error, and no averaging makes a single error normal. So the error distribution is worth checking mainly when the analysis will predict individual responses, when the sample is small, or when a few observations may be of a different kind from the rest.

The errors are never observed, so the first task is to say how well the residuals stand in for them.

21.1.1 Residuals are not the errors

If $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $\mathbf{M}$ is the hat matrix, the residual vector is $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ (Proposition 6.4.3). Write m_{ij} for the entries of $\mathbf{M}$, so that $h_{ii} = m_{ii}$ are the leverages, and c_{ij} for the entries of $\mathbf{I} - \mathbf{M}$. Each residual is a weighted sum of all n errors,

$$\hat{\varepsilon}_i = (1 - h_{ii})\varepsilon_i - \sum_{j \neq i} m_{ij}\varepsilon_j,$$

so even when the errors are independent and identically distributed the residuals are neither. They are correlated, and their variances $\sigma^2(1 - h_{ii})$ differ.

Proposition 21.1.1. What residuals reveal about the errors

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent and identically distributed with mean 0 and variance σ^2 , and let $\text{rank}(\mathbf{X}) = p$.

- (a) $\hat{\varepsilon}_i - \varepsilon_i = -(\mathbf{M}\boldsymbol{\varepsilon})_i$, which has mean 0 and variance $\sigma^2 h_{ii}$. The total $E\|\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon}\|^2$ equals $\sigma^2 p$.
- (b) Let $\hat{\varepsilon}_{(1)} \leq \dots \leq \hat{\varepsilon}_{(n)}$ and $\varepsilon_{(1)} \leq \dots \leq \varepsilon_{(n)}$ be the ordered residuals and ordered errors. Then

$$\frac{1}{n} \sum_{i=1}^n (\hat{\varepsilon}_{(i)} - \varepsilon_{(i)})^2 \leq \frac{1}{n} \|\mathbf{M}\boldsymbol{\varepsilon}\|^2,$$

and the right-hand side has expectation $\sigma^2 p/n$.

- (c) Suppose the errors have skewness $\gamma_1 = E\varepsilon^3/\sigma^3$ and excess kurtosis $\gamma_2 = E\varepsilon^4/\sigma^4 - 3$. If $h_{ii} < 1$, the standardized residual $\hat{\varepsilon}_i/(\sigma\sqrt{1 - h_{ii}})$ has skewness $a_i\gamma_1$ and excess kurtosis $b_i\gamma_2$, where

$$a_i = \frac{\sum_j c_{ij}^3}{(1 - h_{ii})^{3/2}}, \quad b_i = \frac{\sum_j c_{ij}^4}{(1 - h_{ii})^2},$$

with $|a_i| \leq 1$ and $(1 - h_{ii})^2 \leq b_i \leq 1$.

Proof.

- (a) $\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon} = -\mathbf{M}\boldsymbol{\varepsilon}$, with mean zero and covariance $\sigma^2 \mathbf{M}\mathbf{M}^\top = \sigma^2 \mathbf{M}$ (Theorem 2.2.1), whose diagonal is $\sigma^2 h_{ii}$ and whose trace is $\sigma^2 p$.
- (b) Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Pairing the sorted values gives $\sum_i u_{(i)} v_{(i)} \geq \sum_i u_i v_{\pi(i)}$ for every permu-

tation π : if a pairing matches $u_{(i)} < u_{(j)}$ with v -values $b > b'$ respectively, exchanging the two partners changes the sum by $(u_{(j)} - u_{(i)})(b - b') \geq 0$, and finitely many such exchanges reach the sorted pairing. Since sorting does not change $\|\mathbf{u}\|$ or $\|\mathbf{v}\|$, it follows that $\sum_i (u_{(i)} - v_{(i)})^2 \leq \|\mathbf{u} - \mathbf{v}\|^2$. Take $\mathbf{u} = \hat{\boldsymbol{\varepsilon}}$ and $\mathbf{v} = \boldsymbol{\varepsilon}$ and use (a).

- (c) Row i of $\mathbf{I} - \mathbf{M}$ has $\sum_j c_{ij}^2 = (\mathbf{I} - \mathbf{M})_{ii} = 1 - h_{ii}$ because $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent. Expand $\hat{\varepsilon}_i = \sum_j c_{ij}\varepsilon_j$. By independence and $E\varepsilon_j = 0$, every cross term with an index appearing once vanishes, so $E\hat{\varepsilon}_i^3 = E\varepsilon^3 \sum_j c_{ij}^3$ and

$$\begin{aligned} E\hat{\varepsilon}_i^4 &= E\varepsilon^4 \sum_j c_{ij}^4 + 3\sigma^4 \sum_{j \neq k} c_{ij}^2 c_{ik}^2 \\ &= (E\varepsilon^4 - 3\sigma^4) \sum_j c_{ij}^4 + 3\sigma^4(1 - h_{ii})^2. \end{aligned}$$

Dividing by $\sigma^3(1 - h_{ii})^{3/2}$ and $\sigma^4(1 - h_{ii})^2$ gives the skewness and the excess kurtosis. For the bounds, $\max_j c_{ij}^2 \leq \sum_j c_{ij}^2 = 1 - h_{ii}$, so $|\sum_j c_{ij}^3| \leq (1 - h_{ii})^{1/2}(1 - h_{ii})$ and $\sum_j c_{ij}^4 \leq (1 - h_{ii})^2$, while $\sum_j c_{ij}^4 \geq c_{ii}^4 = (1 - h_{ii})^4$. ■

Parts (a) and (b) say that the residuals are good proxies for the errors when p/n is small. Part (b) is the statement that matters for plots, because a normal probability plot shows sorted values: on average the sorted residuals lie within $\sigma\sqrt{p/n}$ of the sorted errors, in root mean square. Part (c) describes what happens when p/n is not small. Each residual is an average of many errors, and averaging pushes a distribution towards normality, so residuals look *more* normal than the errors that produced them. This is sometimes called **supernormality**. The factor b_i measures it exactly.

Example. How much nonnormality survives

Take $n = 30$ observations, an intercept, and $p - 1$ columns of standard normal regressors. Averaged over the observations, the share of the errors' excess kurtosis that survives in the standardized residuals is 0.873 for $p = 2$, 0.467 for $p = 10$ and 0.162 for $p = 20$. The share of skewness is 0.898, 0.532 and 0.162. With centred exponential errors (skewness 2, excess kurtosis 6) and $p = 10$, the residual with the largest leverage, $h_{ii} = 0.57$, has predicted excess kurtosis 1.23, against 1.26 in 200,000 simulated data sets. For the ordered values, the average squared distance between sorted residuals and sorted errors was 0.133, while the bound in (b) averaged 0.330, close to its expectation $p/n = 1/3$.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(2101)
n = 30
designs = {}
for p in (2, 10, 20):
    X = np.column_stack([np.ones(n), rng.normal(size=(n, p - 1))])
    Q, _ = np.linalg.qr(X)
    M = Q @ Q.T # the hat matrix
    designs[p] = (M, np.diag(M).copy())

for p, (M, h) in designs.items():
    # residual i = (row i of C) @ errors
```

```

C = np.eye(n) - M
# excess kurtosis of residual i / that of an error
kurt_kept = (C**4).sum(axis=1) / (1 - h) ** 2
# skewness of residual i / that of an error
skew_kept = (C**3).sum(axis=1) / (1 - h) ** 1.5
print(f"p = {p:2d}: max leverage {h.max():.2f}, "
      f"average share of skewness kept {skew_kept.mean():.3f}, "
      f"of excess kurtosis kept {kurt_kept.mean():.3f}")

```

[Run this code online](#)

21.1.2 Normal probability plots

A **normal probability plot** graphs the sorted values of a sample against the corresponding quantiles of the standard normal distribution. If $U_1, \dots, U_n$ are independent $N(\mu, \tau^2)$, the i th smallest is close to $\mu + \tau q_i$, where q_i approximates the expected i th order statistic of a standard normal sample. A common choice, due to Blom (1958), is

$$q_i = \Phi^{-1}\left(\frac{i - 3/8}{n + 1/4}\right).$$

Normal data give points near a straight line with intercept μ and slope τ . Heavy tails bend both ends away from the line, into an S shape. Skewness bends the plot into a curve that is convex for a long right tail, and isolated points at either end suggest outliers.

For residuals, the unequal variances $\sigma^2(1 - h_{ii})$ should be removed first. The natural choice is the externally studentized residual t_i , which divides $\hat{\varepsilon}_i$ by $s_{(i)}\sqrt{1 - h_{ii}}$, where $s_{(i)}^2$ is the error variance estimate without observation i (Definition 20.1.1). Under normal errors every t_i has the $t(n - p - 1)$ distribution (Theorem 20.3.3), so the t_i share a common marginal law and their sorted values can be compared with one reference. They are not independent, and that makes it hard to judge by eye how much wiggle is normal.

A simulated **envelope** answers that. Under normal errors the vector $(t_1, \dots, t_n)$ is a function of $\hat{\varepsilon}/\|\hat{\varepsilon}\| = (\mathbf{I} - \mathbf{M})\varepsilon/\|(\mathbf{I} - \mathbf{M})\varepsilon\|$ alone, so its joint distribution depends on the design but not on β or σ (Exercise (B1)). We can therefore simulate it exactly: generate normal errors, compute the sorted studentized residuals, repeat, and plot pointwise percentiles (Atkinson 1981). The envelope is pointwise, so a few points outside it are expected even for normal data. A run of consecutive points outside it, or an end of the plot leaving it, is the signal.

Example. Three error laws

Figure 21.1.1 shows plots for one simulated data set of $n = 60$ observations with an intercept and two regressors ($p = 3$), with three error laws of variance one. The normal case stays inside its envelope except for 3 scattered points, and the Shapiro–Wilk test described below gives $p = 0.519$. Errors from a $t(3)$ distribution give the S shape of heavy tails, with the largest residual far above the envelope (Shapiro–Wilk $p < 0.001$). Centred exponential errors give a curved plot whose left end is too short and whose right end is too long ($p = 0.001$).

```

import numpy as np
from scipy import stats

rng = np.random.default_rng(2104)
n, p = 60, 3
X = np.column_stack([np.ones(n), rng.normal(size=(n, 2))])

```

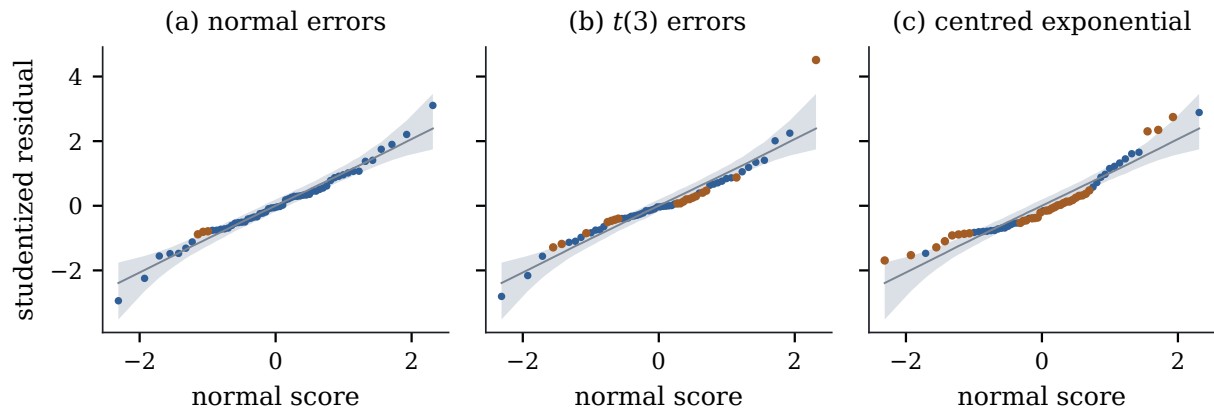


Figure 21.1.1: Normal probability plots of externally studentized residuals, $n = 60$, $p = 3$, with pointwise 95% envelopes from 1999 data sets simulated with normal errors (shaded) and their pointwise median (line). Points outside the envelope are drawn in the second colour. (a) Normal errors. (b) Errors from $t(3)$. (c) Centred exponential errors.

```
Q, _ = np.linalg.qr(X)
h = np.sum(Q**2, axis=1)                                # leverages

def ext_studentized(y):
    """Externally studentized residuals  $t_i = e_i / (s_i \sqrt{1 - h_{ii}})$ ."""
    e = y - Q @ (Q.T @ y)
    s2 = e @ e / (n - p)
    # squared internally studentized residuals
    r2 = e**2 / (s2 * (1 - h))
    return np.sign(e) * np.sqrt(r2 * (n - p - 1) / (n - p - r2))

# plotting positions
blom = stats.norm.ppf((np.arange(1, n + 1) - 0.375) / (n + 0.25))
sims = np.sort([ext_studentized(rng.normal(size=n)) for _ in range(1999)],
                axis=1)
# pointwise envelope
lower, upper = np.percentile(sims, [2.5, 97.5], axis=0)
```

[Run this code online](#)

```
laws = {
    "normal": rng.normal(size=n),
    "t(3)": rng.standard_t(3, size=n) / np.sqrt(3.0),
    "centred exponential": rng.exponential(size=n) - 1.0,
}
beta = np.array([1.0, 2.0, -1.0])
for name, eps in laws.items():
    t = np.sort(ext_studentized(X @ beta + eps))
    outside = np.sum((t < lower) | (t > upper))
    print(f"{name:20s}: {outside} of {n} points outside the envelope, "
          f"Shapiro-Wilk p = {stats.shapiro(t).pvalue:.3f}")
```

[Run this code online](#)

21.1.3 Formal tests

Many tests of normality exist. Four kinds are in common use for residuals.

- The **Shapiro–Wilk** statistic (Shapiro and Wilk 1965) is $W = (\sum_i a_i u_{(i)})^2 / \sum_i (u_i - \bar{u})^2$. The weights a_i come from the means and the covariance matrix of standard normal order statistics, and they make the numerator the square of the best linear unbiased estimate of the standard deviation from the ordered sample, up to a constant. So W compares two estimates of scale, one that is efficient only under normality and one that is always valid. It lies in $(0, 1]$, and small values reject. Its null distribution has no closed form and is computed numerically.
- The **Shapiro–Francia** statistic (Shapiro and Francia 1972) replaces the weights by the plotting positions, which makes it the squared correlation between the sorted sample and the q_i . It is a one-number summary of how straight the normal probability plot is.
- **Moment tests** use the sample skewness $\sqrt{b_1}$ and kurtosis b_2 . The Jarque–Bera statistic $n\{b_1/6 + (b_2 - 3)^2/24\}$ is approximately $\chi^2(2)$ for large normal samples (Jarque and Bera 1980). These tests are directed at the two departures that matter most for inference.
- **Distance tests** compare the empirical and fitted normal distribution functions: Kolmogorov–Smirnov with estimated parameters (Lilliefors 1967) and Anderson–Darling (Anderson and Darling 1952). They are usually less powerful than Shapiro–Wilk.

Applied to residuals, none of these has its nominal null distribution, because the residuals are not a sample. The effect on the level is small. In the design of [Example \(How much nonnormality survives\)](#) with $p = 20$ and normal errors, the 5% Shapiro–Wilk test applied to the raw residuals rejected in 0.045 of 4000 simulated data sets, and applied to the internally studentized residuals in 0.035. The effect on power is not small. Supernormality hides the nonnormality that the test is looking for.

Example. Power lost to supernormality

In the same designs ($n = 30$), the 5% Shapiro–Wilk test applied to the unobservable centred exponential errors rejects in 0.96 to 0.97 of simulated data sets. Applied to the residuals it rejects in 0.92 for $p = 2$, 0.45 for $p = 10$ and only 0.07 for $p = 20$. For errors from $t(5)$, rescaled to variance one, the rates are 0.24 to 0.25 for the errors and 0.23, 0.14 and 0.05 for the residuals.

```
def errors(kind, size, rng):
    """Mean-zero, variance-one errors."""
    if kind == "exp":
        return rng.exponential(size=size) - 1.0
    return rng.standard_t(5, size=size) / np.sqrt(5 / 3)

def power_table(reps, rng):
    """Rejection rates of the 5% Shapiro-Wilk test on errors and on residuals."""
    table = {}
    for kind in ("exp", "t5"):
        for p, (M, h) in designs.items():
            E = errors(kind, (reps, n), rng)
            # residual vectors (M is symmetric)
            R = E - E @ M
            rej_err = np.mean([stats.shapiro(e).pvalue < 0.05 for e in E])
            rej_res = np.mean([stats.shapiro(r).pvalue < 0.05 for r in R])
            table[kind, p] = (rej_err, rej_res)
```

```

print(f"{kind:3s} p = {p:2d}: Shapiro-Wilk rejects "
      f"errors {rej_err:.2f}, "
      f"residuals {rej_res:.2f}")

return table

# a quick version (the book uses 4000 replications)
quick = power_table(400, rng)

```

[Run this code online](#)

A test of normality also answers the wrong question. In small samples it passes error laws that would distort a prediction interval badly; in large ones it rejects departures too small to matter. The useful question is whether the departure is large enough to affect the inference at hand. For coefficient tests without high-leverage points it rarely is. For prediction intervals it can be, as the next result shows.

21.1.4 What nonnormality does to prediction intervals

Under normal errors the interval $\hat{Y}_0 \pm t_{n-p, \alpha/2} s \sqrt{1 + h_0}$ covers a new response $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ with probability exactly $1 - \alpha$, where $h_0 = \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ (Theorem 12.3.1). Without normality its coverage does not tend to $1 - \alpha$ as the sample grows. By Proposition 12.3.3, when $h_0 \rightarrow 0$ and the errors are independent and identically distributed with a continuous distribution function F , the coverage tends to $\Pr(|\varepsilon_0| \leq z\sigma) = F(z\sigma) - F(-z\sigma)$, where $z = \Phi^{-1}(1 - \alpha/2)$, and the probabilities of missing above and below tend to $1 - F(z\sigma)$ and $F(-z\sigma)$.

The limit is the probability that a single error falls within z standard deviations of zero, and it depends on the whole shape of F . Section 12.3 evaluated it for centred exponential and uniform errors at the 80% and 95% levels. Table 21.1.1 extends that comparison to heavy-tailed laws and to the 99% level, and splits the misses between the two tails.

Table 21.1.1. Limiting coverage of the usual prediction interval (Proposition 12.3.3) when the errors are not normal, with the limiting probabilities of missing below and above in parentheses. All five laws have mean zero and variance one.

error law	nominal 80%	nominal 95%	nominal 99%
normal	0.800 (0.100, 0.100)	0.950 (0.025, 0.025)	0.990 (0.005, 0.005)
$t(5)$	0.841 (0.079, 0.079)	0.947 (0.026, 0.026)	0.979 (0.010, 0.010)
$t(3)$	0.887 (0.057, 0.057)	0.957 (0.021, 0.021)	0.979 (0.010, 0.010)
uniform	0.740 (0.130, 0.130)	1.000 (0.000, 0.000)	1.000 (0.000, 0.000)
centred exponential	0.898 (0.000, 0.102)	0.948 (0.000, 0.052)	0.972 (0.000, 0.028)

Heavy tails make a 99% interval too short and an 80% interval too long. The 95% level lies near the crossing point, so checking a 95% interval alone would miss the problem. For the skewed law every miss is above the interval, since a centred exponential error is never below $-\sigma$; at the 99% level it misses above 0.028 of the time, more than five times the nominal 0.005 in that tail.

```

import numpy as np
from scipy import stats

```

```

laws = {
    "normal": stats.norm(),
    "t(5)": stats.t(5, scale=np.sqrt(3 / 5)),
    "t(3)": stats.t(3, scale=np.sqrt(1 / 3)),
    "uniform": stats.uniform(loc=-np.sqrt(3), scale=2 * np.sqrt(3)),
    "centred exponential": stats.expon(loc=-1.0),
}
for name, law in laws.items():
    assert abs(law.var() - 1) < 1e-9 and abs(law.mean()) < 1e-9
    row = []
    for level in (0.80, 0.95, 0.99):
        z = stats.norm.ppf(0.5 + level / 2)
        below, above = law.cdf(-z), law.sf(z)          # miss low, miss high
        row.append(f"{1 - below - above:.3f} ({below:.3f}, {above:.3f})")
    print(f"{name:20s}", " ".join(row))

```

[Run this code online](#)

The remedy is to use the residuals for what they are good at, estimating the error distribution: empirical quantiles of the residuals in place of $\pm z\sigma$ give intervals whose coverage tends to the nominal level, and [Chapter 23](#) refines this by resampling. A transformation of the response ([Chapter 22](#)) can remove skewness at its source.

21.1.5 Exercises

21.1.5.1 A. Check your understanding

Exercise. A1

In simple regression with $n = 5$ and $x = (1, 2, 3, 4, 10)$, compute the leverages and the variances of the five residuals when $\sigma^2 = 1$. Which residual is least like its error, in the sense of [Proposition 21.1.1\(a\)](#)?

21.1.5.2 B. Practice

Exercise. B1

Assume $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. Show that the internally and externally studentized residuals are functions of $\mathbf{g} = \hat{\boldsymbol{\varepsilon}}/\|\hat{\boldsymbol{\varepsilon}}\|$, and that the law of $\mathbf{g}$ does not depend on $\boldsymbol{\beta}$ or σ . Conclude that the simulated envelope of this section is exact, whatever $\boldsymbol{\beta}$ and σ are used in the simulation.

Solution. $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$, and $\boldsymbol{\varepsilon}/\sigma \sim N_n(\mathbf{0}, \mathbf{I})$ whatever $\boldsymbol{\beta}$ and σ are, so $\mathbf{g} = (\mathbf{I} - \mathbf{M})(\boldsymbol{\varepsilon}/\sigma)/\|(\mathbf{I} - \mathbf{M})(\boldsymbol{\varepsilon}/\sigma)\|$ has a fixed law. The internally studentized residual is $r_i = \hat{\varepsilon}_i/(s\sqrt{1 - h_{ii}})$ with $s = \|\hat{\boldsymbol{\varepsilon}}\|/\sqrt{n - p}$, so $r_i = \sqrt{n - p} g_i/\sqrt{1 - h_{ii}}$. The externally studentized residual is the monotone function $t_i = r_i\sqrt{(n - p - 1)/(n - p - r_i^2)}$ of r_i ([Definition 20.1.1](#)). Hence the whole vector of sorted t_i has a law that depends only on $\mathbf{M}$, and it can be simulated with any $\boldsymbol{\beta}$ and σ .

Exercise. B2

For centred exponential errors, $\varepsilon = E - 1$ with E standard exponential, show that the limit in Proposition 12.3.3 at the 95% level is $1 - e^{-2.96}$, and find the probabilities of missing below and above.

Solution. Here $\sigma = 1$ and $z = 1.96$. The event $|E-1| \leq 1.96$ is $E \leq 2.96$, since $E \geq 0 > -0.96$ always. Its probability is $1 - e^{-2.96} = 1 - 0.0518 = 0.948$. The interval never misses below, and it misses above with probability $e^{-2.96} = 0.0518$.

Exercise. B3

In a one-way layout in which every group has m observations, show that the kurtosis factor of Proposition 21.1.1(c) is the same for every residual and equals

$$b = \frac{(1 - 1/m)^4 + (m - 1)/m^4}{(1 - 1/m)^2}.$$

Evaluate it for $m = 2$ and $m = 5$, and explain why residual plots from small groups say little about the tails of the errors.

21.1.5.3 C. Going deeper

Exercise. C1

Under the assumptions of Proposition 21.1.1, suppose $p/n \rightarrow 0$ along a sequence of designs. Let $\hat{F}_n$ and F_n be the empirical distribution functions of the residuals and of the errors. Show that $\hat{F}_n(x) - F_n(x) \rightarrow 0$ in probability at every continuity point x of F .